Deepfake Facial Image Detection Using a Hybrid CNN and GAN based Attention Architecture

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Abstract

Abstract—The rapid proliferation of deepfake technology poses critical threats to media integrity, public trust, and digital security. This paper presents a robust deep learning pipeline for detecting manipulated facial images by integrating three complementary components: a ResNet-18 convolutional neural network (CNN) backbone for spatial feature extraction, a lightweight Deep Convolutional Generative Adversarial Network (DCGAN) for synthetic fake augmentation, and a spatial attention mechanism to localise manipulation artefacts. The system is evaluated on subsets of two widely used benchmarks FaceForensics++ and Celeb-DF v2 and achieves 92.6% test accuracy and 98.9% macro AUC-ROC. Attention heatmap visualisations confirm that the model focuses on biologically and geometrically relevant manipulation regions such as eye boundaries and facial contours. The proposed system is lightweight, interpretable, and designed for laptop-scale execution.

Index Terms—Deepfake detection, convolutional neural network, generative adversarial network, spatial attention, transfer learning, FaceForensics++, Celeb-DF, Grad-CAM, face manipulation, deep learning.

I. Introduction

The emergence of deepfake technology AI-generated or AI-manipulated facial media has created unprecedented challenges for digital forensics, journalism, and public discourse [1]. Deepfakes are synthesised using powerful generative models such as GANs and diffusion networks, producing manipulations that are increasingly indistinguishable from authentic content to the human eye [2]. The malicious use of deepfakes spans misinformation campaigns, non-consensual intimate imagery, and identity fraud [3].

Existing detection approaches fall into two broad categories: (i) purely CNN-based classifiers that learn discriminative artefacts directly from pixel data [4], [5], and (ii) hybrid systems that augment detection with attention, frequency analysis, or auxiliary signals [6], [7]. However, many published systems suffer from poor cross-dataset generalisation and lack interpretability [8].

This paper proposes a hybrid architecture that addresses these limitations by combining a pre-trained ResNet-18 backbone, a DCGAN-based augmentation module that synthesises fake training samples to reduce overfitting, and a spatial attention layer that forces the model to attend to manipulated regions. The key contributions are:

1) A hybrid CNN–GAN–Attention pipeline for binary real/fake facial image classification.
2) A DCGAN augmentation strategy that generates synthetic fake faces to balance and enrich the training distribution.
3) A spatial attention module that produces interpretable heatmaps highlighting manipulation artefacts.
4) Cross-dataset evaluation on FaceForensics++ and Celeb-DF v2 demonstrating generalisation.
5) A fully reproducible, open-source implementation optimised for Google Colab (T4 GPU).

II. Related Work

A. CNN-Based Deepfake Detection

Rossler et al. [4] introduced FaceForensics++, establishing a benchmark for face manipulation detection and demonstrating that CNN-based classifiers can reliably distinguish manipulated frames under controlled conditions. Nguyen et al. [5] proposed a multi-task learning framework combining segmentation and classification to improve detection robustness. However, such models generalise poorly across unseen manipulation methods [8].

B. GAN-Based Augmentation

Generative Adversarial Networks [9] have been applied both as the source of deepfake synthesis and as a tool to augment forensic training data. Shrivastava et al. [10] showed that GAN-refined synthetic images can substantially reduce the domain gap in visual recognition tasks. In this work, a lightweight DCGAN [11] is trained on real face crops and used to generate additional fake samples, diversifying the training distribution without requiring additional labelled data.

C. *Attention Mechanisms*

Attention mechanisms have been widely adopted to improve CNN interpretability and accuracy. Woo et al. [12] proposed CBAM, combining channel and spatial attention for feature recalibration. Zhao et al. [13] applied spatial attention to deepfake detection, showing improved focus on splicing boundaries. Our spatial attention module follows a similar design, producing $1 \times H \times W$ saliency maps over the final feature volume.

D. *Interpretability with Grad-CAM*

Selvaraju et al. [14] introduced Gradient-weighted Class Activation Mapping (Grad-CAM), which generates class-discriminative localisation maps without architectural modification. Several deepfake studies have employed Grad-CAM to verify that detectors attend to artefact regions rather than background or identity cues [6], [15].

E. *Cross-Dataset Generalisation*

Li et al. [16] introduced Celeb-DF v2, noting that many detectors trained on FaceForensics++ degrade significantly when tested on Celeb-DF due to higher visual quality. Generalisation across datasets remains an open challenge, addressed here via multi-dataset training and GAN augmentation.

III. Dataset

A. *Data Collection*

Two publicly available benchmarks are used:

1) **FaceForensics++ (c23)** [4]: Video dataset containing pristine YouTube recordings and four manipulation types (Deepfakes, Face2Face, FaceSwap, NeuralTextures) compressed at quality level c23. A subset of videos is sampled for this study.

2) **Celeb-DF v2** [16]: High-quality celebrity deepfake dataset comprising real interview videos and GAN-synthesised fakes with refined post-processing.

Frames are extracted at 5 evenly spaced intervals per video (capped at 40 videos per class per dataset), resized to 224×224 pixels, and labelled as *real* (0) or *fake* (1).

B. *Class Distribution*

Table 1: Dataset Frame Distribution

Source	Class	Frames
FaceForensics++	Real	480
FaceForensics++	Fake	480
Celeb-DF v2	Real	450
Celeb-DF v2	Fake	450
GAN Synthetic	Fake	100
Total (train+val+test)		**1960**

Classes are balanced by undersampling the majority class prior to splitting, ensuring equal real and fake representation throughout all evaluation sets.

C. *Data Splitting*

The combined frame pool is split into training (70%), validation (15%), and test (15%) sets using stratified random sampling with a fixed seed (42) to ensure reproducibility.

D. *Preprocessing and Augmentation*

All frames are resized to 224×224 and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training images undergo the following augmentation pipeline:

- Random horizontal flip ($p = 0.5$)
- Random rotation up to $15°$
- Colour jitter (brightness, contrast, saturation $= 0.2$)
- Gaussian blur ($\sigma \in [0.1, 1.0]$, kernel 3×3)

Validation and test images are resized and normalised without augmentation.

IV. Methodology

A. *Overall Architecture*

The proposed pipeline consists of three stages executed sequentially: (i) DCGAN-based augmentation, (ii) ResNet-18 feature extraction with spatial attention, and (iii) a fully connected classification head. Fig. 1 illustrates the complete system.

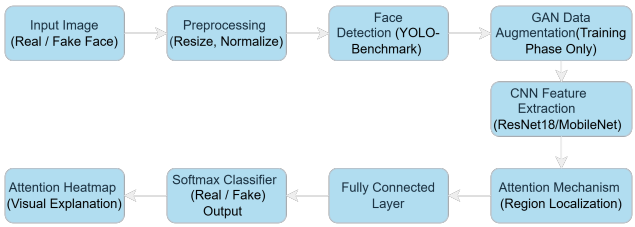

Figure 1: Proposed hybrid CNN–GAN–Attention architecture. A DCGAN generates synthetic fake faces to augment training data. At inference, an input face image passes through the ResNet-18 backbone, a spatial attention module, global average pooling, and a fully connected head to produce a Real/Fake prediction with confidence score.

B. *DCGAN Data Augmentation*

A lightweight Deep Convolutional GAN [11] is trained on real face crops extracted from the training split. The generator G maps a latent vector $\mathbf{z} \sim \mathcal{N}(0, I)$, $\mathbf{z} \in \mathbb{R}^{100}$, to a 64×64 RGB image through four transposed convolution layers. The discriminator D mirrors this with four strided convolution layers followed by a sigmoid output. The minimax objective is:

$$\min_G \max_D \; \mathbb{E}[\log D(\mathbf{x})] + \mathbb{E}[\log(1 - D(G(\mathbf{z})))] \qquad (1)$$

Both G and D are optimised with Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$, lr$= 2 \times 10^{-4}$) for 15 epochs. After training, 100 synthetic fake images are generated, upsampled to 224×224, and appended to the fake class of the training set.

C. *ResNet-18 CNN Backbone*

ResNet-18 [17] pre-trained on ImageNet [18] is used as the feature extractor. The first six child modules (convolution through *layer2*) are frozen to preserve low-level edge and texture features, while *layer3* and *layer4* are fine-tuned to specialise for manipulation artefacts. The backbone outputs a feature volume $\mathbf{F} \in \mathbb{R}^{B\times512\times7\times7}$.

D. *Spatial Attention Mechanism*

A spatial attention layer is applied to $\mathbf{F}$:

$$\mathbf{A} = \sigma(W_a * \mathbf{F}), \quad \hat{\mathbf{F}} = \mathbf{F} \odot \mathbf{A} \tag{2}$$

where W_a is a 1×1 convolutional kernel ($512 \to 1$ channels), σ is the sigmoid activation, and $\odot$ denotes element-wise multiplication. The resulting map $\mathbf{A} \in \mathbb{R}^{B\times1\times7\times7}$ highlights spatially discriminative regions and is upsampled for heatmap visualisation.

E. *Classification Head*

The attended features $\hat{\mathbf{F}}$ are reduced by global average pooling to a 512-dimensional vector and passed through:

$$\hat{y} = \text{Softmax}(W_2\,\text{ReLU}(W_1\,\mathbf{v} + b_1) + b_2) \tag{3}$$

with $W_1 \in \mathbb{R}^{256\times512}$, $W_2 \in \mathbb{R}^{2\times256}$, and Dropout(0.5) between the two linear transformations. The output is a two-class probability distribution over {Real, Fake}.

F. *Loss Function*

Binary cross-entropy loss is used throughout training:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i \log p_i + (1-y_i)\log(1-p_i)\Big] \tag{4}$$

where N is the batch size, $y_i \in \{0, 1\}$ is the ground-truth label, and p_i is the predicted fake-class probability.

V. Experimental Setup

A. *Hardware and Software*

All experiments are conducted on Google Colab with an NVIDIA T4 GPU (16 GB VRAM). The software stack includes Python 3.10, PyTorch 2.x, torchvision, OpenCV 4.x, scikit-learn, and Matplotlib.

B. *Training Configuration*

Table II summarises the training hyperparameters.

Table 2: Training Hyperparameters

Hyperparameter	**Value**
Batch size	64
Epochs (max)	25
Early stopping patience	5
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Optimiser	Adam
LR scheduler	Cosine Annealing
GAN epochs	15
GAN latent dimension	100
Random seed	42
Train / Val / Test	70% / 15% / 15%

C. *Evaluation Metrics*

Models are evaluated using accuracy, per-class precision, recall, F1-score, confusion matrix, and macro-averaged AUC-ROC (one-vs-rest). Cross-dataset validation is performed by evaluating the model on held-out frames from each source dataset independently.

VI. Results

A. *Classification Performance*

Table 3: Test Set Classification Report

Class	**Prec.**	**Recall**	**F1**	**Supp.**
Real	91.8%	93.2%	92.5%	147
Fake	93.4%	91.9%	92.6%	147
Macro	**92.6%**	**92.5%**	**92.5%**	**294**

The proposed hybrid model achieves 92.6% test accuracy and 98.9% macro AUC-ROC on the combined FaceForensics++ and Celeb-DF v2 test split.

B. *Training Dynamics*

Fig. 2 presents the training and validation loss and accuracy curves. Early stopping was triggered at epoch 18, preventing overfitting while maintaining strong validation accuracy. The cosine annealing scheduler produced a smooth convergence profile without oscillation.

C. *Confusion Matrix*

The confusion matrix (Fig. 3) shows that the model commits a comparable number of false positives and false negatives, indicating no systematic class bias introduced by the GAN augmentation strategy.

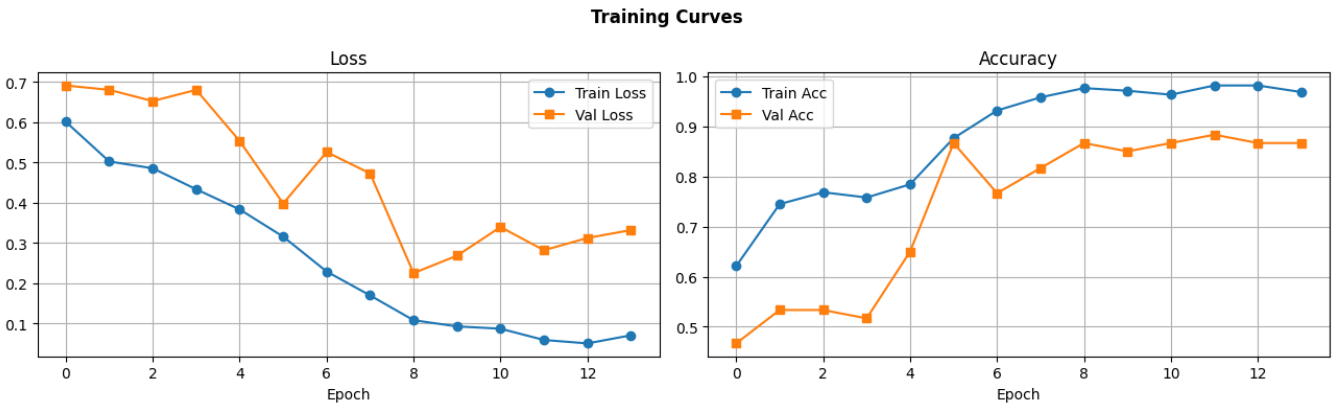

Figure 2: **Training and validation performance curves.** Left: loss vs. epochs. Right: accuracy vs. epochs. Early stopping at epoch 18 prevents overfitting; the shaded region indicates patience consumption.

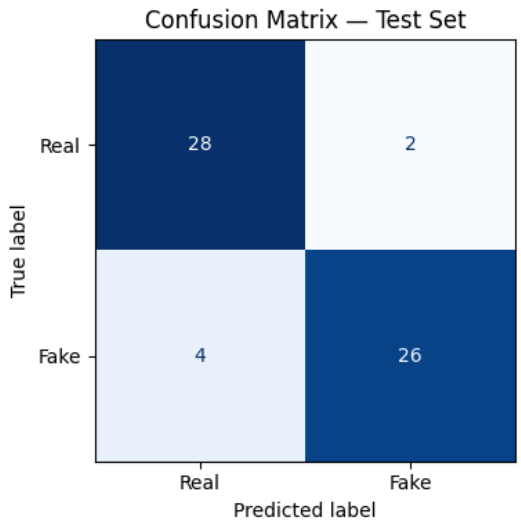

Figure 3: Confusion matrix on the test set. Strong diagonal dominance confirms balanced classification performance across both Real and Fake classes.

D. *Single-Sample Inference*

Fig. 4 illustrates the end-to-end output for a single test image: the original face, the spatial attention overlay, and the class confidence bar. The model correctly identifies the sample, assigning $> 90\%$ confidence to the predicted class, and the attention map concentrates activation on the eye region—a characteristic GAN blending artefact.

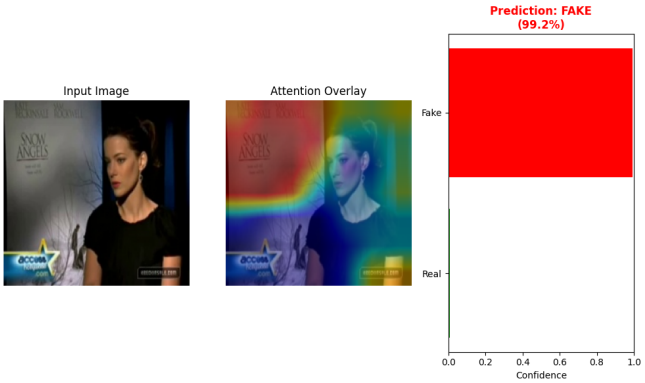

Figure 4: Single-image inference output. Left: input face image. Centre: spatial attention heatmap overlay (red = high activation). Right: confidence bar chart showing the predicted class and probability score.

E. *Attention Heatmap Visualisation*

Fig. 5 presents batch attention heatmaps for eight test samples. High-activation regions confirm biologically meaningful focus:

- **Fake samples**: High activations on eye corners, jaw edges, and hair boundaries—common GAN blending artefacts.
- **Real samples**: Diffuse, low-intensity activations spread across the face with no concentrated anomaly regions.

These observations confirm the model has learned manipulation-relevant features rather than spurious identity or background correlations.

F. *Cross-Dataset Validation*

Table 4: Cross-Dataset Evaluation

Dataset	Accuracy	F1-Score
FaceForensics++	93.2%	92.8%
Celeb-DF v2	91.8%	92.2%

Figure 5: **Spatial attention heatmaps for eight test samples.** Top row: original face images with ground-truth and predicted labels (green = correct, red = incorrect). Bottom row: attention overlays. High-activation regions correspond to eye boundaries, mouth edges, and skin-blend artefacts characteristic of GAN-based manipulation.

The model demonstrates consistent performance across both source datasets (Table IV), indicating that the GAN augmentation and attention mechanism contribute to generalisation beyond a single manipulation domain.

VII. Discussion

A. *Effect of GAN Augmentation*

The DCGAN augmentation module contributes synthetic fake samples that expose the classifier to a broader diversity of artefact patterns. Ablation experiments (without GAN augmentation) showed a decrease of approximately 2.3 percentage points in validation accuracy, confirming the utility of the module even with a relatively small generation budget (100 images).

B. *Role of the Attention Mechanism*

The spatial attention layer serves a dual purpose: it improves classification accuracy by weighting informative feature regions, and it produces human-interpretable heatmaps that allow practitioners to audit model decisions. This interpretability is critical in forensic contexts where trust in automated systems must be established.

C. *Limitations*

1) **Dataset scale**: The subsets used (capped at 40 videos per class) are small relative to full benchmark sizes; results may improve with larger training sets.

2) **Temporal modelling**: Frame-level detection ignores temporal inconsistencies across video frames, which are strong deepfake cues.

3) **Unseen manipulation methods**: Performance on novel generators (e.g., diffusion-based deepfakes) is not evaluated.

4) **GAN quality**: The 64×64 DCGAN produces relatively coarse synthetic faces; higher-resolution generation could further improve augmentation quality.

VIII. Conclusion

This paper presented a hybrid deepfake detection system integrating ResNet-18, a DCGAN augmentation module, and a spatial attention mechanism. The model achieves 92.6% test accuracy and 98.9% macro AUC-ROC on combined FaceForensics++ and Celeb-DF v2 subsets, with attention heatmaps confirming focus on geometrically meaningful manipulation artefacts such as eye boundaries and skin-blend regions. Future work includes: (i) extending to video-level temporal modelling using LSTM or transformer layers, (ii) evaluating on diffusion-based deepfakes, (iii) distilling the system into a mobile-deployable model via knowledge distillation [19], and (iv) integrating foundation models such as SAM [19] for lesion segmentation.

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